

National University of Computer and Emerging Sciences



Programming Fundamentals CS188 Laboratory Manual

Course Instructor

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Section

BDS-1A1 & A2

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FAST School of Computing

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	Lab No 4			
	Course Name:	Programming Fundamentals	Course Code:	CS 1002
	Program:	BS(DS)	Semester:	Fall 2021
	Duration:	2.5 hours	Total Points:	10 + 40 + 50
	Lab Date:	Friday, October 15, 2021	Weight	4%
	Section:	BDS-1A	Page(s):	

Instruction/Notes: Cheating during the lab will result in negative marks

Topics Covered: Repetition Structure

Use any of the IDEs including CodeBlocks, Visual Code, Visual Studio or the simple IDE available at <https://www.onlinegdb.com/> to write the programs

Submission Instructions:

1. Save all .cpp files according to the following naming convention
{ROLLNO}_{ACTIVITYNO}_{TASKNO}.cpp **FOR EXAMPLE.** 21L-XXXX_A01_P01.cpp, 21L-XXXX_A01_P02.cpp, 21L-XXXX_A02_P01.cpp
2. Now create a new folder according to the following naming convention
{ROLLNO}_{LABNO} e.g. 21L-XXXX_L02
3. Move all of your .cpp files to this newly created directory and compress it into a single file.
4. Submit this compressed file on Google Classroom.
5. You will get 10 Bonus point if you will follow these instructions correctly

Activity No 1

Problem No 1:

The following C++ code segments can be used to print **n** spaces or **n** * on the output device/stream

Code segment to print spaces
<pre>for(int i = 0; i < n; i++) cout<<" ";</pre>
Code Segment to print *
<pre>for(int i = 0; i < n; i++) cout<<"* ";</pre>

Use these code segments within a loop to create programs that can print the following patterns

Filled Square of height H The value of H will be given by the user	Sample output for H = 8 <pre>* *</pre>
Hollow Square of height H The value of H will be given by the user	Sample output for H = 6 <pre>* * * * * * * * * * * * * * * * * *</pre>

Filled Square of
height **H**

The value of H will
be given by the
user

Sample output for H = 8

```

                *
              * *
            * * *
          * * * *
        * * * * *
      * * * * *
    * * * * *
  * * * * *
* * * * *
```

Hollow Triangle of
height **H**

The value of H will
be given by the
user

Sample output for H = 8

```

                *
              * *
            *   *
          *     *
        *       *
      *         *
    *           *
  *             *
*               *
* * * * * * * *
```

Activity No 2.

Problem No 1:

Write a C++ program to prints the first **n** terms of the following **Tick-Tock** series. The value of n is taken as input.

The first 8 terms of the **Tick-Tock** series, i.e., for **n=8** are as follows:

$$2 - 5 + 4 - 10 + 6 - 15 + 8 - 20$$

Can you guess the pattern?

The odd terms are multiples of 2 and in increasing order (2, 4, 6, 8 etc.). The even terms are increasing multiples of 5 and in negative form. The terms are alternatively positive and negative. Nothing should be printed if the value of n is below 1.

For Example:

If the input is n=5, the program must print: **2 - 5 + 4 - 10 + 6**

If input is n=10, the program must print: **2 - 5 + 4 - 10 + 6 - 15 + 8 - 20 + 10 - 25**

Problem No 2:

Write a C++ program to prints the first **n** terms of the following Fibonacci series. The value of n is taken as input.

The first 10 terms of the **Fibonacci series**, i.e., for **n = 10** are as follows:

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55$$

So the first two terms are 1 and 1 and every successive term is sum of the previous two terms.

For Example:

If the input is n=5, the program must print: **1, 1, 2, 3, 5.**

If input is n=10, the program must print: **1, 1, 2, 3, 5, 8, 13, 21, 34, 55.**